Multilingual Topic Modeling of Sri Lankan Parliamentary Debates: An NLP Pipeline for Trilingual Hansard Analysis

D.H.N. Dhanapala
himathd.23@cse.mrt.ac.lk
Department of Computer Science & Engineering, University of Moratuwa
Moratuwa, Sri Lanka

K.S.H. Daishika
harend.23@cse.mrt.ac.lk
Department of Computer Science & Engineering, University of Moratuwa
Moratuwa, Sri Lanka

H.A. Kuruppu
himandhik.23@cse.mrt.ac.lk
Department of Computer Science & Engineering, University of Moratuwa
Moratuwa, Sri Lanka

N.A.K. Nawagamuwa
ashinik.23@cse.mrt.ac.lk
Department of Computer Science & Engineering, University of Moratuwa
Moratuwa, Sri Lanka

N.S. Seneviratne
sithijas.23@cse.mrt.ac.lk
Department of Computer Science & Engineering, University of Moratuwa
Moratuwa, Sri Lanka

Abstract

This study presents a multilingual natural language processing (NLP) framework for analyzing Sri Lankan parliamentary debates (Hansards), a trilingual corpus of speeches in Sinhala, Tamil, and English, including code-mixed content. Since the Hansard data is distributed in PDF format, reliable text extraction is difficult due to layout complexity, multilingual scripts, and OCR-related noise. To address this, we employ a large language model (LLM)-based extraction approach using Gemini, which improves the reconstruction of structured parliamentary text. The extracted corpus is processed through a topic modeling pipeline that combines BGE-M3 multilingual embeddings, UMAP dimensionality reduction, and HDBSCAN clustering, followed by hierarchical aggregation to derive interpretable macro-topics. In addition, a hybrid semantic–lexical BiTopic framework is explored as an extension for improving interpretability and recovering speeches otherwise classified as noise. Results show that traditional LDA fails to produce coherent multilingual topics, whereas the proposed embedding-based pipeline successfully identifies cross-lingual thematic structure across major policy domains. The final pipeline recovers 30 interpretable macro-topics whose temporal patterns align with major national events such as the 2019 Easter Sunday attacks and the 2022 economic crisis. These findings demonstrate the effectiveness of combining LLM-assisted extraction with multilingual topic modeling, providing a strong foundation for downstream analysis of parliamentary discourse.

Keywords

multilingual NLP, topic modeling, BERTopic, parliamentary debates, Hansard, Sinhala, Tamil, HDBSCAN, semantic embeddings, clustering

1 Introduction

The Parliament of Sri Lanka serves as the principal national forum for deliberation on governance, public policy, economic planning, and social welfare. Its official transcripts, known as Hansards, constitute one of the richest publicly available records of Sri Lankan political discourse. Published in Sinhala, Tamil, and English, these transcripts preserve legislative debate, policy argumentation, and inter-party exchange in their original linguistic form. As such, they provide a valuable resource for understanding policy priorities, ideological positioning, and the evolution of parliamentary attention across major national issues.

The systematic analysis of this corpus has both scholarly and civic value. Topic modeling of parliamentary debates can reveal how legislative attention is distributed across policy domains, track the rise and decline of issues over time, and provide a structured basis for downstream tasks such as sentiment analysis, political stance detection, and representative accountability studies. However, these benefits are difficult to realize using standard natural language processing (NLP) pipelines without substantial adaptation.

The Sri Lankan Hansard corpus is distinguished by three interrelated characteristics. First, it is inherently trilingual: speeches may be delivered in Sinhala, Tamil, English, or code-mixed combinations of these languages, sometimes even within a single speaker turn. Second, Sinhala and Tamil exhibit highly agglutinative morphology, causing a single lexical root to appear in many surface forms and reducing the effectiveness of traditional bag-of-words representations. Third, the source documents are distributed primarily as PDF files containing dual-column layouts, mixed-script rendering, and formatting irregularities that limit the reliability of conventional OCR-based extraction.

These challenges motivate the development of a multilingual NLP framework specifically tailored to Sri Lankan parliamentary discourse. This study proposes an end-to-end pipeline that combines LLM-based text extraction with multilingual embeddings and clustering-based topic modeling to generate coherent thematic structure from noisy and linguistically diverse debate data. In addition to a BERTopic-based pipeline, a hybrid semantic–lexical architecture, BiTopic, is explored as an experimental extension for improving topic interpretability and reducing embedding blur.

This work is guided by three research questions. First, how can a topic modeling pipeline be designed to operate effectively over a large-scale trilingual corpus containing Sinhala, Tamil, and English, including code-mixed text, while preserving cross-lingual semantic coherence? Second, can macro-level thematic groupings be derived empirically from the hierarchical structure of discovered micro-topics rather than specified arbitrarily? Third, does augmenting dense semantic embeddings with lexical information improve topic boundary sharpness relative to a purely embedding-based baseline?

The main contributions of this paper are summarized as follows:

- An end-to-end multilingual topic modeling pipeline for Sri Lankan Hansard data, integrating LLM-based text extraction, multilingual embedding generation, dimensionality reduction, density-based clustering, and topic representation.
- A systematic evaluation of multilingual embedding models for clustering long-form, code-mixed parliamentary speeches using cross-lingual semantic similarity, semantic retrieval, and anisotropy analysis.
- A comparative analysis of clustering algorithms, highlighting the precision–coverage trade-off and demonstrating the suitability of HDBSCAN for high-purity topic extraction in multilingual parliamentary discourse.
- A hybrid semantic–lexical BiTopic framework that combines dense embeddings with lexical grounding to improve topic interpretability and recover semantically relevant speeches that are otherwise classified as noise.
- A foundation for downstream parliamentary analytics, including sentiment analysis, stance detection, and broader studies of legislative attention over time.

2 Related Work

Computational analysis of parliamentary debates has been widely explored using corpora such as EuroParl, the British Hansard, and the U.S. Congressional Record. These studies have supported applications such as machine translation, agenda tracking, sentiment analysis, and discourse analysis. However, most prior work has focused on high-resource, predominantly monolingual corpora, unlike Sri Lankan parliamentary data, which combines Sinhala, Tamil, and English in a trilingual and often code-mixed setting.

Traditional topic modeling in political text has been dominated by probabilistic methods such as Latent Dirichlet Allocation (LDA). While effective for monolingual corpora, LDA relies on bag-of-words representations and word co-occurrence patterns, limiting its ability to capture contextual meaning and cross-lingual semantic equivalence. These weaknesses become more severe in agglutinative languages such as Sinhala and Tamil, where a single lexical root may appear in many surface forms. Alternative sparse methods such as Non-negative Matrix Factorisation share similar limitations.

Recent work has shifted toward neural and embedding-based approaches that operate over dense semantic representations. Contextualised Topic Models and related neural methods have demonstrated improved coherence compared to classical probabilistic models, but they have not been systematically evaluated on corpora that combine low-resource languages, agglutinative morphology, and dense code-mixing. BERTopic forms the primary methodological foundation of this work. It combines contextual embedding, dimensionality reduction, density-based clustering, and class-based TF-IDF keyword extraction. This modular design is particularly suitable for parliamentary discourse because it does not require the number of topics to be specified in advance and can assign weakly related or procedural speeches to noise rather than forcing them into clusters.

The effectiveness of such models depends strongly on the underlying multilingual embeddings. Models such as multilingual-e5, BGE-M3, LaBSE, and paraphrase-multilingual-mpnet-base-v2 have been proposed for cross-lingual semantic representation. Multilingual-e5 achieves strong semantic similarity and retrieval performance, but its embedding space may exhibit high anisotropy, which can negatively affect clustering. In contrast, BGE-M3 provides a more evenly distributed embedding space together with long-context support, making it more suitable for clustering long-form parliamentary speeches.

For topic discovery, density-based clustering methods are especially relevant because parliamentary corpora contain both substantive policy speeches and procedural interventions. HDBSCAN can identify clusters of varying density while assigning outlier speeches to noise, whereas partitioning methods such as K-Means force all observations into clusters and may reduce topic purity. For clustering evaluation, metrics such as ARI and NMI provide global measures of agreement, while B-Cubed metrics better capture document-level clustering quality in imbalanced corpora.

Despite these developments, robust topic modeling frameworks for low-resource, code-mixed parliamentary corpora remain limited. Prior work has not addressed multilingual topic modeling for the Sri Lankan Hansard corpus at scale, nor explored hybrid semantic–lexical topic modeling specifically for improving interpretability and recovering speeches that would otherwise remain unclustered. This study addresses these gaps through a multilingual topic modeling pipeline and an experimental BiTopic framework tailored to Sri Lankan parliamentary discourse.

3 Dataset

3.1 Data Collection

The dataset used in this study consists of Sri Lankan parliamentary debate transcripts (Hansards), obtained from the official website of the Parliament of Sri Lanka. The Hansard serves as the verbatim public record of parliamentary proceedings and is published as PDF documents within the parliamentary archive. Each document typically corresponds to a single day of parliamentary sitting and contains speeches, procedural annotations, speaker identifiers, and multilingual content reflecting the language used by each Member of Parliament.

To construct the corpus, a dedicated web scraping pipeline was developed to systematically collect Hansard PDF documents from the archive. The scraper traversed year-level index pages, identified linked PDF files for each parliamentary session, and downloaded them into a structured local directory organized by year and sitting date. The collection period spans from 2017 to 2026, covering both a pre-crisis baseline period and the politically and economically turbulent years surrounding the 2019 Easter Sunday attacks, the 2022 economic crisis, the Aragalaya uprising, and subsequent IMF-linked economic restructuring.

Documents that did not contain the main debate speeches targeted by the topic modeling pipeline, such as non-sitting records, question sessions, and committee reports, were excluded from the primary collection. The resulting corpus focuses on substantive parliamentary debate content in Sinhala, Tamil, and English.

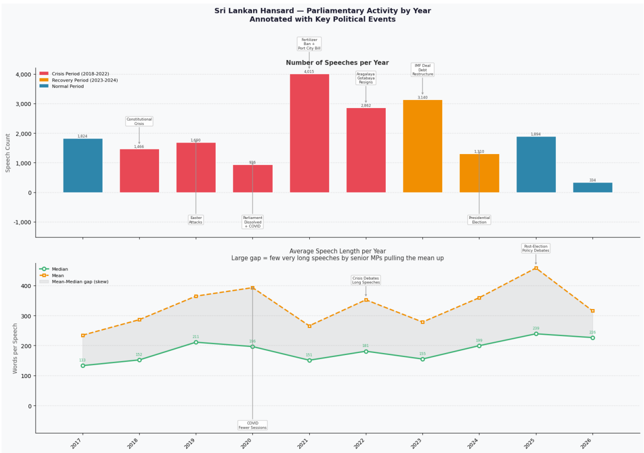

Figure 1: Parliamentary Activity by Year Annotated with Key Political Events.

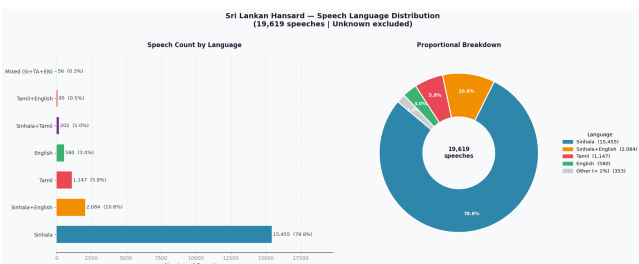

Figure 2: Speech Language Distribution.

3.2 LLM-Based Extraction

Initial text extraction from the collected Hansard PDFs was performed using Tesseract OCR. However, the complex structure of the source documents, including dual-column layouts, multilingual content, and font inconsistencies, resulted in substantial noise and structural errors in the extracted output. These issues limited the reliability of the text and reduced its suitability for downstream NLP tasks.

To overcome these limitations, a large language model-based extraction pipeline was adopted using Google Gemini. A carefully engineered prompt was designed to perform three tasks simultaneously: extract only substantive speeches delivered by identified Members of Parliament, suppress procedural and non-informative content such as interruptions, Speaker rulings, and short utterances, and enforce a structured output format in which each speech is represented as a single coherent text segment attributed to a normalized speaker identifier.

A key design decision in this process was to preserve multilingual and code-mixed text in its original form, without translation, transliteration, or linguistic normalization. This ensured that Sinhala and Tamil content remained in its original script and morphological form, preserving linguistic properties that are themselves analytically significant. Compared to the OCR baseline, the Gemini-based pipeline produced substantially cleaner output by resolving dual-column interleaving, reproducing mixed-script content more accurately, and maintaining speaker attribution more consistently.

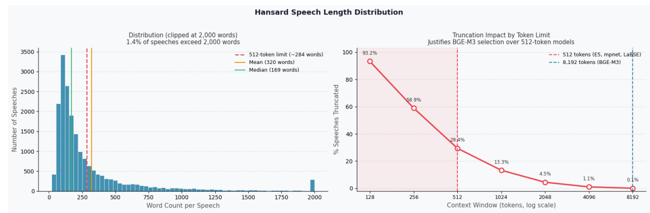

Figure 3: Hansard Speech Length Distribution.

3.3 Ground Topic Evaluation Subset

To enable quantitative evaluation of the topic modeling pipeline under controlled conditions, a manually curated subset of 300 parliamentary speeches was annotated with expert-assigned topic labels. The annotation process used a taxonomy of 16 thematic categories, including Economy & Finance, Governance & Legal Reform, Parliamentary Affairs, Infrastructure & Energy, Health & Social Welfare, National Security & Public Order, Agriculture & Fisheries, Education, Foreign Affairs & Diplomacy, Reconciliation & Transitional Justice, Labour & Migration, Culture & Religion, Disaster Management, Technology & Innovation, Tourism Promotion, and Environmental Protection.

Language identification on the annotated subset revealed 273 Sinhala speeches (91.0%), 19 Tamil speeches (6.3%), and 8 English speeches (2.7%). Prior to clustering evaluation, topic labels with fewer than five speech instances were excluded to reduce the effect of severe class imbalance, yielding a filtered subset of 288 speeches across 13 topic categories.

3.4 Trilingual Stopword Engineering

A custom stopword list was developed specifically for the Sri Lankan parliamentary domain. Standard English stopword lists are insufficient for this corpus, as they do not account for Sinhala and Tamil function words, nor for parliamentary vocabulary that appears frequently across Hansard records regardless of topic.

The stopword construction process was carried out in three stages. First, language-specific base lists were compiled for Sinhala, Tamil, and English. Second, parliamentary procedural vocabulary was added across all three languages, including formulaic address terms, speaker acknowledgements, procedural motions, and recurring administrative expressions. Third, corpus-level frequency analysis was performed on the full speech collection, and terms occurring in more than 80% of speeches were added to the list. The final stopword list comprises 1,045 terms across Sinhala, Tamil, and English and was used in lexical stages of the pipeline to suppress non-informative vocabulary while preserving semantically meaningful content.

4 Methodology

4.1 Baseline Topic Modeling using LDA

Before adopting embedding-based approaches, a baseline topic modeling experiment was conducted using Latent Dirichlet Allocation (LDA). To ensure a fair comparison for the multilingual dataset, the LDA pipeline was adapted to handle Sinhala and Tamil text by incorporating a custom whitespace tokenizer and the curated trilingual stopword list. Standard preprocessing techniques such as stemming and lemmatization were deliberately avoided because they often distort non-Latin scripts.

Despite these adaptations, LDA failed to produce coherent and unified topics for the parliamentary corpus. Its reliance on word co-occurrence limited its ability to capture cross-lingual semantic relationships, resulting in language-specific topic fragmentation rather than consistent thematic groupings. Furthermore, the morphological characteristics of Sinhala and Tamil led to multiple surface variations of the same root word being treated as separate features, thereby diluting topic distributions. These limitations motivated the transition to embedding-based topic modeling methods.

4.2 Embedding Model Selection

The proposed topic modeling pipeline depends strongly on the quality of semantic embeddings. Sri Lankan parliamentary debates present challenges such as multilingual content (Sinhala, Tamil, and English), domain-specific political vocabulary, and long-form speech structure. These characteristics require embedding models that preserve cross-lingual semantic relationships while maintaining contextual integrity over long text sequences. To address this, multiple multilingual embedding models were evaluated using a custom benchmark designed for parliamentary discourse.

Four embedding models were evaluated on a 300-speech ground-truth subset using the following methods:

- **Method 1 - Cross-Lingual Semantic Similarity (STS):** Mean cosine similarity was computed between speech pairs across different languages within the same topic. This measures whether semantically equivalent speeches are mapped to similar regions of the embedding space regardless of language.
- **Method 2 - Semantic Retrieval:** English queries related to parliamentary topics were issued against the corpus, and retrieval effectiveness was measured using Precision@5 and Mean Reciprocal Rank (MRR). This reflects the model's ability to support cross-lingual information retrieval.
- **Method 3 - Anisotropy Analysis:** Global anisotropy was measured using the mean off-diagonal cosine similarity of randomly sampled speech embeddings. Highly anisotropic spaces indicate vector collapse, which reduces the effectiveness of downstream dimensionality reduction and clustering.

The evaluated models include multilingual-e5-large-instruct, BAAI/bge-m3, sentence-transformers/LaBSE, and paraphrase-multilingual-mpnet-base-v2.

- **Multilingual-e5-large-instruct:** This model achieved the strongest cross-lingual semantic similarity and retrieval performance, showing strong alignment across Sinhala, Tamil, and English. However, it exhibited high anisotropy, causing embeddings to collapse into a narrow vector space that negatively affects clustering. Its 512-token context limit also truncates long parliamentary speeches.
- **bge-m3:** This model provided the most balanced performance for clustering. Although its cross-lingual similarity scores were lower than E5, it produced a well-distributed embedding space with lower anisotropy, enabling clearer topic separation. Its 8,192-token context window also allows full speeches to be encoded without truncation.
- **sentence-transformers/LaBSE:** LaBSE is designed for cross-lingual alignment and showed stable semantic similarity performance. However, it achieved lower retrieval effectiveness and is limited by a 512-token context window.
- **paraphrase-multilingual-mpnet-base-v2:** This model served as a general-purpose multilingual baseline. While it produced relatively stable embeddings with lower anisotropy, its cross-lingual semantic performance was weaker and it remained limited by a 512-token context window.

Table 2: Clustering-based embedding evaluation on 300 Hansard speeches.

Model	Dim	Time (s)	Topics	Outlier %	Silhouette	Davies–Bouldin ↓	Coherence c_v
LaBSE	768	268	3	0.28	0.90	3.10	0.46
mpnet-base-v2	768	171	16	18.97	0.79	3.09	0.25
BAAI/bge-m3	**1024**	**1449**	**89**	**27.57**	**0.81**	**2.48**	**0.38**
multilingual-e5-large	1024	2334	70	38.63	0.71	3.03	0.04

Embedding Model selection - Although multilingual-e5-large-instruct achieved the strongest cross-lingual semantic alignment, its high anisotropy and limited context capacity reduce its suitability for clustering-based topic modeling. The collapsed nature of its embedding space weakens cluster separability, which is critical for downstream dimensionality reduction and density-based clustering.

In contrast, BAAI/bge-m3 offers a better balance between semantic quality and embedding space structure. Its lower anisotropy produces a more uniformly distributed vector space, improving UMAP-based dimensionality reduction and HDBSCAN clustering. Its extended context window also preserves full parliamentary speeches without truncation.

Therefore, BAAI/bge-m3 was selected as the embedding model for the proposed topic modeling pipeline.

4.3 Dimensionality Reduction

Following embedding generation using BGE-M3, Uniform Manifold Approximation and Projection (UMAP) was applied to reduce the high-dimensional embedding space to a lower-dimensional form suitable for clustering. The original 1024-dimensional embeddings were projected into a 5-dimensional space using UMAP with $n\_neighbors = 15$, $n\_components = 5$, $min\_dist = 0.0$, and cosine distance as the similarity metric.

This step is important for improving density-based clustering. UMAP preserves local neighbourhood relationships while reshaping the embedding space to produce clearer density variations, enabling HDBSCAN to identify coherent clusters more effectively.

Table 1: Embedding model benchmark results on 300 Hansard speeches.

Model	STS Mean ↑	Retrieval MRR ↑	P@5 ↑	Anisotropy ↓	Verdict
multilingual-e5-large-instruct	0.827	0.810	0.56	0.909	✗ Collapsed
BAAI/bge-m3	**0.523**	**0.516**	**0.68**	**0.552**	**✓ Selected**
LaBSE	0.487	0.324	0.72	0.492	✓ Healthy
paraphrase-multilingual-mpnet-base-v2	0.424	0.536	0.76	0.496	✓ Healthy

The low *min_dist* value further encourages tighter grouping of semantically similar points, improving cluster compactness and separation.

A separate 2-dimensional UMAP projection (*min_dist* $= 0.1$) was also generated for visualization, providing an interpretable view of the embedding structure.

4.4 Clustering Algorithm Comparison

Four clustering algorithms were evaluated on UMAP-reduced embeddings using the 300-speech ground-truth subset. To reduce the effect of class imbalance, labels with fewer than five speeches were excluded, yielding a filtered dataset of 288 speeches across 13 topic categories.

The algorithms were evaluated using B-Cubed Precision, Recall, and F1-score, together with Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and Silhouette Score. For density-based methods such as HDBSCAN and OPTICS, noise points (label -1) were treated as singleton clusters in B-Cubed computation, ensuring a fair comparison with partition-based methods that assign all speeches to clusters.

The results show a clear trade-off between cluster purity and coverage. Partition-based methods, including K-Means and Agglomerative (Ward) clustering, assign all speeches into a fixed number of clusters ($K = 13$), leading to higher B-Cubed F1 scores (0.324 and 0.340, respectively) due to full coverage. Agglomerative clustering performed slightly better in terms of label recovery.

In contrast, HDBSCAN achieved much higher cluster purity (B-Cubed Precision $= 0.673$) and stronger geometric separation (Silhouette $= 0.532$) by excluding low-density or ambiguous points, with 41.7% of speeches classified as noise. This shows that HDBSCAN prioritizes compact and well-defined clusters over exhaustive coverage. OPTICS, while also density-based, performed less effectively due to higher noise rejection and over-segmentation, producing more clusters than the ground-truth distribution.

Clustering Method selection - Based on the evaluation, HDBSCAN was selected as the clustering algorithm for this study due to its ability to adapt to variable-density structures, capture non-linear cluster boundaries, and effectively isolate noise. When combined with UMAP-based dimensionality reduction, it enables the extraction of semantically coherent and well-separated topics.

Although HDBSCAN yields a lower B-Cubed F1 score compared to partition-based methods, this reflects a deliberate trade-off favoring cluster purity over coverage. For the intended application, downstream sentiment analysis requires clearly defined and semantically consistent topics, as mixed-topic clusters can lead to ambiguous or misleading sentiment signals. The speeches identified as noise predominantly correspond to procedural content or weakly related discourse, and their exclusion contributes to improved topic quality and interpretability.

4.5 Final BERTopic Pipeline

The final topic modeling pipeline integrates embedding generation, dimensionality reduction, clustering, and topic representation within the BERTopic framework. BGE-M3 was used as the embedding model, UMAP as the dimensionality reduction method, and HDBSCAN as the clustering algorithm. A CountVectorizer with the custom trilingual stopword list was used to retain meaningful lexical features from Sinhala, Tamil, and English while suppressing boilerplate procedural vocabulary. Topic representations were generated using class-based TF-IDF together with KeyBERT-inspired and Maximal Marginal Relevance techniques to improve keyword diversity and interpretability.

Applied to the full corpus, the BERTopic pipeline produced 336 micro-topics and assigned 6,921 speeches (35.4%) to noise, retaining 12,632 speeches in substantive topic clusters. This micro-topic layer provided the basis for the macro-topic aggregation procedure described below.

4.6 Macro-Topic Aggregation

The 336 micro-topics produced by HDBSCAN were analytically useful at fine granularity but too numerous for higher-level interpretation. To derive a stable and interpretable topic taxonomy, a two-stage macro-topic aggregation procedure was introduced.

First, micro-topics were ranked by size and filtered using a Pareto-style threshold so that only dominant clusters accounting for the majority of substantive speech mass were retained. This step reduced the influence of hyper-local or highly sparse clusters.

Second, centroids of the retained micro-topics were computed in the BGE-M3 embedding space and subjected to agglomerative hierarchical clustering using cosine distance and average linkage. Rather than fixing the number of macro-topics arbitrarily, the resulting dendrogram was cut at an empirically determined distance threshold corresponding to the most prominent structural gap within an interpretable range. This procedure yielded a final partition of 30 macro-topics, which formed the main thematic output of the study.

4.7 Experimental BiTopic Framework

In addition to the primary BERTopic pipeline, a BiTopic-based framework was explored as an experimental extension. BiTopic combines semantic and lexical similarity before topic formation, allowing both contextual meaning and explicit token-level evidence to contribute within a unified clustering framework.

Table 3: Clustering algorithm benchmark results (288 speeches, 13 true topics, $K\_true = 13$)

Algorithm	K Found	Noise %	BCubed P	BCubed F1	ARI	NMI	Silhouette
K-Means	13	0.0%	0.349	0.324	0.184	0.379	0.388
Agglomerative (Ward)	13	0.0%	0.355	0.340	0.191	0.398	0.377
HDBSCAN	**11**	**41.7%**	**0.673**	**0.313**	**0.104**	**0.340**	**0.532**
OPTICS	19	36.1%	0.670	0.250	0.073	0.351	0.497

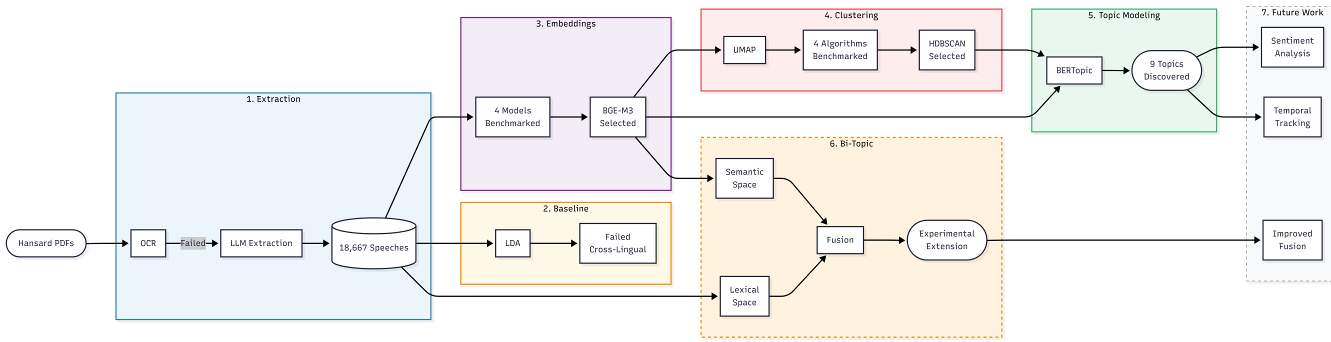

Figure 4: System architecture and NLP pipeline for multilingual Hansard analysis.

For each speech, two complementary representations were constructed: a dense semantic embedding generated using BGE-M3 and a sparse lexical vector generated using CountVectorizer. Pairwise cosine similarity was computed independently in the semantic and lexical spaces, normalized to a common scale, and combined through weighted fusion to produce a final similarity matrix, which was then converted into a distance matrix for clustering.

The fused similarity used an asymmetric weighting scheme, with semantic similarity given a dominant weight and lexical similarity a smaller regularizing weight. This reflects a meaning-first strategy while preserving enough lexical structure to avoid conflating semantically adjacent but institutionally distinct speeches. During experimentation, some speeches classified as noise by the BERTopic pipeline were assigned to meaningful topics under the BiTopic framework, suggesting that lexical grounding can improve topic interpretability and recover sparsely distributed but thematically relevant speeches.

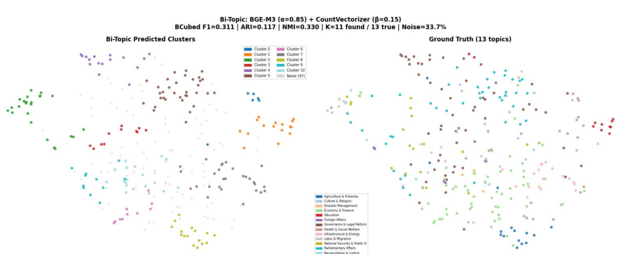

Figure 5: BiTopic Framework for Multilingual Topic Modeling.

5 Results

5.1 Macro-Topic Discovery

The final pipeline was applied to the full corpus of 19,553 speeches and produced a data-driven taxonomy of 30 interpretable macro-topics. These topics span major parliamentary policy domains, including Energy Economy & Fuel Crisis, Agriculture & Plantation, Education, Healthcare, Constitutional Reform, Parliamentary Oversight, Human Rights & International Relations, Fisheries, Tourism, and Easter Sunday Attacks & National Security.

The resulting topic structure is both semantically coherent and politically interpretable. Economy-related topics dominate the corpus, reflecting Sri Lanka's sustained fiscal and energy crises during the study period. Institutional and constitutional topics capture repeated debates around executive power, electoral reform, and public oversight. At the same time, narrower topics such as SAITM Medical Education, Cricket Administration, and Dairy Industry demonstrate the pipeline's ability to isolate politically meaningful niche debates.

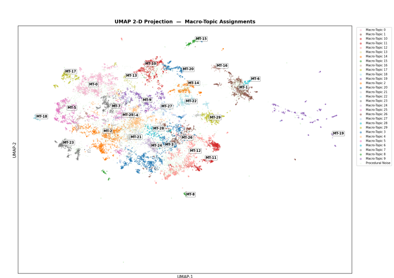

Figure 6: Macro Topic Assignments

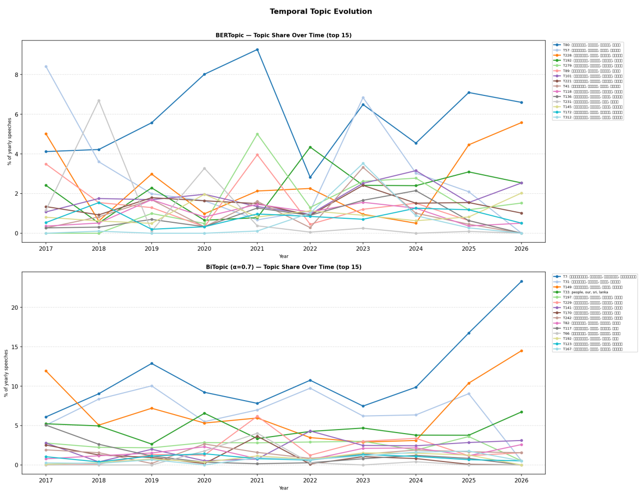

Figure 8: Temporal Topic Evolution.

Table 4: Discovered topics on Hansard speeches.

Topic ID	Speeches	Interpreted Label	Key Terms
MT-0	509	Parliamentary Procedure & Standing Orders	ප්‍රශ්න, ආණ්ඩුක්‍රම, ස්ථාවර නියෝග, පාර්ලිමේන්තු, ව්‍යවස්ථා
MT-1	1,395	Energy Economy & Fuel Crisis	රුපියල්, ණය, ආර්ථික, මෙගාවොට්, තෙල් ටැංකි, සූර්ය
MT-2	738	Tamil-Medium Parliamentary Debates	இவ்வாறு, நடவடிக்கைகளை, பிரதமர், பாதிக்கப்பட்ட
MT-3	340	Infrastructure — Ports, Aviation & Urban Dev.	වරාය නගරය, හම්බන්තොට, ශ්‍රීලන්කන්, ගුවන් යානා
MT-4	736	Agriculture, Plantation & Price Policy	කිලෝවක් රුපියල්, තේ, පොහොර, ගම්මිරිස්, කාබනික
MT-5	852	Social Welfare, Housing & Water	නිවාස, ජල, සමෘද්ධි, ආපදා, ගංවතුර
MT-6	424	Parliamentary Oversight & Public Accounts (COPE)	රජයේ ගිණුම්, COPE, බැඳුම්කර, කාර්ය සාධන
MT-7	724	Education — Schools, Universities & Teachers	පාසල්, ගුරු, විශ්ව විද්‍යාල, විභාග, අධ්‍යාපනය
MT-8	568	Constitutional Amendments & Executive Governance	ආණ්ඩුක්‍රම ව්‍යවස්ථා, සංශෝධනය, ජනාධිපති, රාජාසන
MT-9	360	Parliamentary Privilege & Conduct	රාමනායක, රන්ජන්, වරප්‍රසාද, ආචාර ධර්ම
MT-10	227	Cooperative Finance & Microfinance	සමුපකාර, ක්ෂුද්‍ර මූල්‍ය, බැංකු, තැන්පතු
MT-11	685	Healthcare, Pharmaceuticals & NMRA	nmra, ඖෂධ, වෛද්‍ය, රෝහල්, කොවිඩ්
MT-12	89	Utilities Legislation & Media Law	ලංකා විදුලිබල, ජනමාධ්‍ය, රූපවාහිනී, සංශෝධන
MT-13	317	English-Medium Governance & Peace Discourse	peace, majority, sovereignty, constitution, tamil
MT-14	653	Environment, Land & Wildlife Conservation	වනජීවී, ඉඩම්, අලි-මිනිස් ගැටුම, සංරක්ෂණ
MT-15	481	Parliamentary Condolences & Obituaries	නිවන් සුව, අභාවය, ශෝකය, ජ මු
MT-16	184	Specific Parliamentary Events (Political)	අමරකීර්ති, Commonwealth, Queen Elizabeth
MT-17	439	Easter Sunday Attacks & National Security	පාස්කු ප්‍රහාරය, සහරාන්, කොමිෂන් වාර්තාව, ත්‍රස්ත
MT-18	370	Transport & Road Infrastructure	දුම්රිය, බස් රථ, අධිවේගී, මාර්ග සංවර්ධන
MT-19	234	Foreign Employment, Tourism & Diaspora	විදේශ රැකියා, සංචාරකයන්, ශ්‍රමිකයන්, ප්‍රේෂණ
MT-20	189	Human Rights & International Relations	මානව හිමිකම්, ජිනීවා, ත්‍රස්තවාදය, ඇමෙරිකා
MT-21	90	Dairy Industry & Livestock	කිරි පිටි, දියර කිරි, පශු සම්පත්, ලීටර්
MT-22	114	SAITM Medical Education & Defence University	සයිටම්, SAITM, කොතලාවල, නිදහස් අධ්‍යාපනය
MT-23	389	Electoral Reform & Local Government Elections	සීමා නිර්ණය, මැතිවරණ කොමිෂන්, ඡන්ද, මනාප
MT-24	148	Tax Policy — Excise, Gems & Revenue	සුරා බදු, මැණික්, VAT, ස්වර්ණාභරණ
MT-25	240	Cricket & Sports Administration	ලංකා ක්‍රිකට්, ICC, ක්‍රීඩකයන්, ක්‍රිකට් ආයතනය
MT-26	563	Judicial Affairs, Crime & Anti-Corruption	නඩු, අධිකරණ, දූෂණ, මහේන්ද්‍රන්, රිමාන්ඩ්
MT-27	133	Fisheries & Maritime Industry	ධීවර, මත්ස්‍ය, යාත්‍රා, බහුදින, ඩීසල්
MT-28	155	Plantation (Estate) Sector & Labour	වතු, වැටුප, කම්කරු, තේ වතු, තොණ්ඩමන්
MT-29	286	Heritage, Women, Children & Municipal Waste	පුරාවිද්‍යා, කසළ, කාන්තාවන්, ළමා, බුද්ධ ශාසන

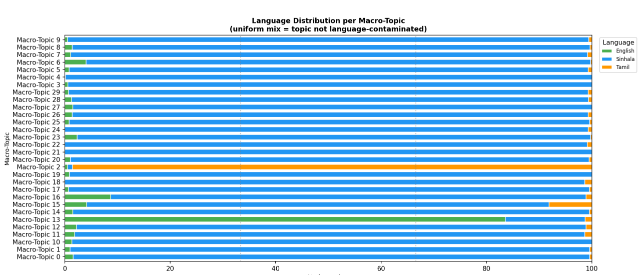

Figure 7: Speech Language Distribution.

5.2 Temporal and Cross-Lingual Patterns

Temporal trajectories of the discovered topics align closely with major national events. Economy-linked topics show clear surges during the 2022 economic crisis and Aragalaya period, while the Easter Sunday Attacks topic rises sharply around 2019 and the subsequent period of parliamentary scrutiny. Healthcare-related topics also show distinct prominence during the COVID-19 period.

The language distribution across macro-topics further shows that most topics contain speeches from more than one language. While a small number of topics are more strongly associated with Tamil-medium or English-medium debate, the majority of topics reflect clustering by thematic content rather than by language identity.

5.3 Clustering Quality

Quantitative evaluation shows that the embedding-based pipeline outperforms traditional bag-of-words baselines. HDBSCAN achieved the strongest cluster purity and geometric separation among the clustering methods tested, although it also assigned a substantial fraction of speeches to noise. The embedding benchmark similarly showed that BGE-M3 provided a more suitable representation space for clustering than the alternative models evaluated, balancing semantic performance with lower anisotropy and long-context support.

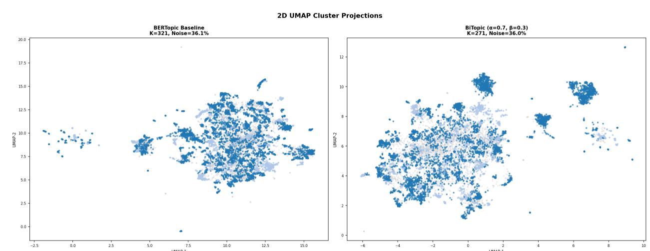

Figure 9: 2D UMAP Cluster Projections.

5.4 BiTopic Results

The BiTopic framework was evaluated as a secondary experiment alongside the primary BERTopic-based pipeline. The results indicate that incorporating lexical similarity alongside dense semantic embeddings can improve topic boundary sharpness and recover some speeches that would otherwise remain unclustered. However, this also increased the overall noise rate in some configurations, indicating a trade-off between interpretability and coverage.

6 Discussion

The results show that multilingual parliamentary debate can be modeled effectively using an embedding-based topic modeling pipeline. The 30 discovered macro-topics provide a meaningful representation of Sri Lankan parliamentary discourse and recover both broad policy domains and narrower issue-specific debates without explicit supervision.

A key observation is the external validity of the topic structure. The temporal trajectories of several topics align with major national events, including the 2019 Easter Sunday attacks, the COVID-19 period, and the 2022 economic crisis. This suggests that the model captures genuine shifts in legislative attention rather than simply grouping speeches by repeated vocabulary.

The cross-lingual behavior of the final topic structure is also important. Most macro-topics contain speeches from more than one language, indicating that clustering is driven primarily by thematic content rather than language identity. At the same time, a few topics remain more strongly associated with Tamil-medium or English-medium debate, reflecting real differences in parliamentary language use.

Another important finding is the role of macro-topic aggregation in improving interpretability. While the initial micro-topic layer captures fine-grained distinctions, it is too fragmented for higher-level analysis. Aggregating these into macro-topics produces a more stable and analytically useful representation of the corpus.

The BiTopic experiment further suggests that hybrid semantic–lexical modeling is a useful extension for recovering thematically relevant speeches that are difficult to cluster using embeddings alone. However, the increase in fragmentation and noise under some fusion settings indicates that lexical grounding must be introduced carefully. As a result, the semantic BERTopic pipeline remained the stronger main model, while BiTopic is best viewed as an exploratory direction for future refinement.

7 Limitations

This study has certain limitations. First, the supervised evaluation relies on a small ground-truth subset of 300 speeches, which is heavily skewed towards Sinhala. Second, topics with fewer than 5 speeches were filtered out prior to evaluation, leading to their exclusion from the clustering evaluation. This reflects the genuine imbalance of parliamentary time allocation across topics rather than a sampling artifact. Third, HDBSCAN's noise rejection rate of 41.7% on the benchmark corpus means a substantial fraction of parliamentary time is excluded from the final topic model. While this is appropriate for sentiment analysis applications, it represents a loss of coverage that may be unacceptable for other research questions requiring complete corpus analysis.

8 Future Work

This topic modeling framework establishes a foundation for further analysis of Sri Lankan parliamentary discourse, particularly in sentiment analysis. Future work will focus on two main directions.

1. Aspect-Based Sentiment Analysis (ABSA): with coherent, cross-lingual topic clusters identified, targeted sentiment models can be applied to analyze how political actors respond to specific policy issues. For example, examining sentiment within the Economic Crisis cluster can provide insights into shifting political priorities and legislative responses.

2. Temporal Sentiment Tracking: Analyzing how sentiments and topics shift over different parliamentary sessions to understand the dynamic nature of political discourse.

9 Conclusion

This study presented a multilingual topic modeling framework for Sri Lankan parliamentary debates, combining LLM-based text extraction, multilingual embeddings, density-based clustering, and hierarchical aggregation to recover thematic structure from a noisy trilingual corpus. The proposed pipeline was shown to overcome the main limitations of traditional bag-of-words approaches, particularly in relation to cross-lingual semantic fragmentation, agglutinative morphology, and long-form parliamentary text.

The results demonstrate that the combination of BGE-M3, UMAP, and HDBSCAN provides a robust basis for extracting semantically coherent and politically interpretable topics from Sri Lankan Hansards. The final macro-topic aggregation procedure yielded 30 interpretable thematic domains whose temporal patterns align with major political and economic events, confirming the substantive validity of the recovered topic structure. The exploratory BiTopic extension further showed that lexical grounding can improve topic boundary sharpness and recover otherwise difficult speeches, though with a trade-off in coverage.

Overall, this work establishes a strong foundation for multilingual parliamentary NLP in Sri Lanka and provides a structured basis for downstream tasks such as sentiment analysis, stance detection, and longitudinal analysis of legislative attention.

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